

A Peek Into Tripwire, The Open Source Security Tool

With the growing demand for cyber security in the open source world, it has become important to know about all the methods and tools available. Many available tools can be used manually or scripted to automate security-related tasks. With this article, we start a mini-series about such tools, covering small, useful utilities for open source systems that help systems and network administrators strengthen their IT infrastructure.



As we all know, one never ever achieves 100 per cent cyber security; rather, it is a continuous process of improvement. In the past issues, I covered cyber attacks in detail, which laid the ground for understanding the fundamentals of security. We learnt that there is no single utility, commercial or open source, which can fix all the vulnerabilities and secure IT infrastructure from attacks. We also understood that implementing an intrusion detection system is imperative, besides the rest that's already in place. For each vulnerability or attack vector, there are multiple utilities available to help network administrators implement, monitor and audit cyber security. Many famous tools such as Nmap, Chkrootkit, Snort, etc, have already been covered. This series introduces you to tools that are not commonly known. The first in the series is Tripwire.

Tripwire

In the continuously changing IT infrastructure scenario, it is important for a systems administrator to know if and when a file structure is changed. Such a change can stem out of a regular code release, or can be a malicious file update caused by viruses, spy-ware or root-kits. These call for detailed file update monitoring and logging. Tripwire is an open source tool available for all Linux distros, for just this purpose. It detects and reports any unauthorised changes to files and directories. Now let's

understand how it works, and how to install and use it effectively.

At a high level, Tripwire works in a 'sample and compare' way. When Tripwire is installed and run for the first time, it scans the entire file system to create a database containing information about the file structure, as well as the date and time stamp and other vital parameters for each file. System binaries as well as user application files are scanned, and their information stored in the database. At this point, Tripwire enters into monitoring mode, and watches every file access. Each update is compared with the reference database, and intelligent actions or alerts are generated out of it. Tripwire monitors and detects file additions, deletions, updates, etc, and also watches for file attribute changes. If the changes are legitimate, the administrator can update the Tripwire database to accept these changes. The contents of this baseline database are indeed supposed to be secure. Tripwire asks for a site key, which is a sort of password that must be entered by the administrator while installing the tool for the first time. Though this is an optional step, I definitely recommend that you do so. This site key encrypts the database, and decrypts it on the fly, while performing a comparison with file-system scan results. During the installation phase, the site key resides on the machine itself in a plain-text form, but after the installation is complete, it is removed, prompting the installer or administrator to remember it, just like a password.